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CPE 101 — Past Questions & Materials
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Course: CPE 101
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 100

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Paper 1 of 1 — CPE 101 G. O. Uzedhe Course Handout

Level 100 • First Semester • Handout

BASIC ELECTRONICS

(CPE 101)

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Objectives

The objectives of the course are to:

- Explain the historical background of electronics.
- Describe the principle of operation of electronic devices.
- Explain the behavior of electronic devices in relation to application area.
- Describe the role of electronics in computer design and application.
- Explain the functions of AC and DC signals in a computer system.
- Explain the advent of semiconductors and their growth in electronic device development.
- Describe the process of semiconductor operation.
- Explain the use of semiconductors in logic operations
- Illustrate semiconductor logic operation for computer system design.

Learning Outcomes

At the end of the course, the student should be able to:

- 1. discuss the principles and operation of basic electronic devices.
- 2. describe the origin, formation, and application of electronics.
- 3. describe functions of basic electronic devices in a computer circuit.
- 4. distinguish between AC and DC electronic device operations,
- 5. recognize and analyze the function of an electronic device in a circuit;
- 6. explain the development of semiconductor devices.
- 7. identify the use of semiconductor electronic devices for logic operations.
- 8. write down the various logic functions of semiconductor device.

Course Content

History and overview of electronics: from vacuum tubes to large scale integration, including reasons for studying electronics, selected important areas of application, role of electronics in computer engineering. DC Circuit Analysis: Kirchorff's Law, Mesh and Nodal equations, Superposition Theorem, Thevenins Theorem, Nortons Theorem, Maximum Power Transfer, Transients (RL and RC) Circuits. AC Circuit Analysis: Alternating current, voltage, frequency, phase

angle; maximum, rms and average values of waveforms.

Inductive and capacitive reactances; power in ac circuits;

use of complex algebra in the solution of ac circuits;

Resonance.

Course Content

Semi-conductivity: materials and properties, electrons and holes, concept of doping, acceptors and donors, p and n-type materials, conductivity and resistivity. Diodes and Circuits:

symbol and representation, diode operation and

characteristics, region of operation and limitations, zener and

schottky diodes, diode circuit and load line, diode application

in rectifier and dc/dc converter, diode logic functions - AND

and OR . Bipolar Junction Transistors (BJT): physical

structure of BJT, symbol and circuit representation, NPN and

PNP transistor operation, voltage-current characteristics of

transistors, transistor region of operation and limitation,

transistor circuit analysis, biasing for logic application,

Resources

- Lecture Note
- Basic Electronics by Dr. Sanjay Sharma
- Principles of Electronic Devices and Circuits by B. L. Theraja and Dr. R. S. Sedha

History and overview of electronics

Electronics:

Electronics is the science and technology of the motion of charges in a gas, vacuum or semiconductor. A historical division was made in the beginning of the twentieth century that separated between the already existing field of electrical engineering and the emerging field of electronics engineering.

Where electrical engineering dealt only with devices that are depended solely on the motion of electrons in metals such as motors, generators, light bulbs, and wire communication systems (telephone and telegraph).

At the end of the twentieth century the historical division between electrical and electronic engineering no longer serves its original function. The distinction between information processing and energy processing serves to separate electronics from the rest of electrical engineering.

The nature of the discipline of electronics comprises of three fields: communication, computation, and control.

History and overview of electronics

The history of electronics is divided into two main parts: the vacuum-tube era and the transistor era that started at 1948.

The Discovery of Vacuum Tubes: Following the discovering of the electron tube by Ferdinand Braun in 1897, Fleming in 1904 invented the vacuum tube diode referred to as “valve”. It consisted of a heated wire, the filament and was separated by a short distance from a metallic plate. De Forest put a third electrode (grid) into the Fleming valve and invented a triode tube referred to as “audion”. The audion serve as the first amplifier. These inventions led to early radio and television technologies in vacuum tube radios and cathode-ray tube imaging devices.

History and overview of electronics

History and overview of electronics

The transistor:

The PN junction diode was the first semiconductor electronic components to be invented by Russel Ohl. This invention of the PN junction diode became the basic building block for semiconductor technology development. Diodes

In 1947, Brattain and Bardeen invented the point-contact transistor at the Bell Labs and Shockley later discovered the junction transistor in 1950. Bell Labs grew single crystals of germanium. From this development, the first grown- junction transistors were made and were followed year later alloy-junction transistors.

Transistors

History and overview of electronics

The Field Effect Transistor

Much of the work of the bipolar transistor studies the effect of electric field, Shockley proposed the junction field effect transistor (JFET) in 1951. In 1958, the first JFET was produced in France. The techniques in JFETs development led to the Metal Oxide Semiconductor Field Effect Transistor (MOSFET). The structure consists of a metallic electrode (the gate) on the SiO₂ between two electrodes (source and drain).

In 1960, Bell Laboratories reported the first MOSFET device.

History and overview of electronics

Integrated Circuit (IC):

Kilby in 1958 built both an oscillator and a multivibrator from germanium, making circuit interconnections from thermally bonded gold wire. The real keys to IC manufacture were the planer and batch processing. The planer process used transistors in which the base and emitter regions were diffused into the collector. Batch processing permitted many IC "chips", to be

History and overview of electronics

By 1961, Texas Instruments was producing ICs commercially, followed soon afterward by other companies. The following approximate dates give some indication of the increasing component count per chip: 1951—discrete transistors, 1960—small-scale integration (SSI) with fewer than 100 components, 1966—medium scale integration (MSI) with 100 to 1000 Components. With the development of the Complementary Metal Oxide Semiconductor (CMOS), a circuit configuration that employ both (PMOS) and (NMOS) devices, circuit density became higher at less energy consumption.

History and overview of electronics

The development led to Large Scale Integration (LSI) with 1000 to 9,999 components. Most MOS ICs produced in 1970 went into the production of calculators. Several ICs manufacturers proposed calculator architectures into its circuit functions. These computational devices led to the development of the Microprocessor first developed at Intel in 1969 and 4-bit microprocessors were introduced in 1971. As the complexity increases with reduced components size, Very Large Scale Integrated (VLSI) circuits were introduced containing 10,000 to 99,999 components. Further development have since then led to circuits containing 100,000 to millions of components in Ultra Large Scale Integration (ULSI) technology.

History and overview of electronics

APPLICATIONS of

Electronics

Electronic components:

- Capacitor
- Cathode ray tube (CTR)
- Diode
- Field effect transistor (FET)
- Integrated circuit (IC)
- Junction gate field effect transistor (JFET)

- Inductor

- Liquid crystal display

History and overview of electronics

- Light dependent resistor (LDR)

- Light emitting diode (LED)

- MOSFET transistor

- BJT transistors

- Resistors

- Relays

- Switches

- Thermistor, and more.

History and overview of electronics

Consumer Electronics:

Include products like –

- Audio Systems—Radios, Home theaters, etc.
- Video Systems—TV (Television), Computer, Laptop, Digital Camera,
- DVD Players
- Home and Kitchen Appliances
- GPS
- Mobiles Phones etc.

History and overview of electronics

Communication:

Electronic communication systems

connect people around the world. Using telephones, Internet and computers, people in different countries

communicate almost instantly. Radios transmit sounds and televisions transmit sounds and pictures great distances. Cellular telephones enable a person to call another person. Within seconds, fax machines send and receive copies of documents over telephone lines/Satellite.

History and overview of electronics

Information processing:

Scientists, artists, students, government and business workers, and hobbyists at home all rely on computers, Internet to handle huge amounts of information quickly and accurately. Computers solve difficult mathematical problems, maintain vast amounts of data, create complex simulations, and perform a multitude of other tasks that help people in their everyday lives.

History and overview of electronics

Medicine and research:

Include product like X-ray machines
ECG (Electrocardiogram) use radiation to take images of bones and internal organs. Radiation therapy, or radiotherapy, uses X-rays and other forms of radiation to fight cancer.

Many hearing-impaired people depend on hearing aids to electrically amplify sound waves.

History and overview of electronics

Computers:

Computers and other electronic instruments provide scientists and other researchers with powerful tools to better understand their area of study.

Computers, for example, help scientists design new drug molecules, track weather systems, and test theories about how galaxies and stars develop.

Electron microscopes use electrons rather than visible light to magnify specimens 1 million times or more.

History and overview of electronics

Automation:

Electronic components enable many common home appliances, such as refrigerators, washing machines, and toasters, to function smoothly and efficiently. People can electronically program coffeemakers, lawn sprinklers, and many other products to turn on and off automatically.

Microwave ovens heat food quickly by penetrating it with short radio waves produced by a vacuum tube.

History and overview of electronics

Instrumentation:

Measuring Instruments like CRO, Multimeter, ph-meter, strain gauge, Frequency Counter are used in different

Laboratory/organisations. Many automobiles have electronic controls in their engines and fuel systems. Electronic devices also control air bags, which inflate to protect a driver and passengers in a collision.

Lecture 2 : DC Circuit Analysis

- Charge and Current • Thevenins Theorem
- Kirchorff's Law • Nortons Theorem
- Mesh and Nodal • Maximum Power equations Transfer
- Superposition • Transients (RL and Theorem RC) Circuits

DC Circuit Analysis

- Charge and Current

Charge is an attractive force denoted by the letter Q and has units of coulombs.

Electrons are negatively charged and protons are positively charged.

All electrons and protons exhibit the same magnitude of charge, roughly coulombs.

Thus, one coulomb is equivalent to the charge exhibited by approximately $1/1.6 \times 10^{-19}$, or electrons.

Further, opposite charges attract while like charges repel, similar to the poles of a magnet.

It is possible to move charge from one point to another. The rate of charge movement over time is called current. It is denoted by the letter I and has units of amperes (or amps).

DC Circuit Analysis

One amp of current is defined as one coulomb per second.

$$1 \text{ amp} \equiv 1 \text{ coulomb} / 1 \text{ second}$$

That is, one amp can be visualized as approximately electrons passing through a wire in a period of one second.

$$I = Q/t$$

Where

I = is the current in amps,

Q = is the charge in coulombs,

t = is the time in seconds.

DC Circuit Analysis

• Example 1:

In the course of one-half second, a certain battery delivers a charge of three coulombs. Determine the resulting current.

DC Circuit Analysis

Solution:

Class Exercise:

A device delivers a current of 25 mA.

Determine the charge

transferred in two

seconds along with the

equivalent total number To find the total number of electrons transferred, of electrons. one coulomb is equivalent to (6.242×10^{18}) electrons.

Total electrons = $Q \times$ number of electrons per coulomb

$$= 50\text{mC} \times 6.242 \times 10^{18}$$

$$= 3.121 \times 10^{17} \text{ electrons}$$

DC Circuit Analysis

Energy and Voltage:

Energy is defined as the ability to do work.

It is denoted by the letter W . The basic unit is the joule although other units are sometimes used (for example, the calorie or the kilowatt-hour, kWh).

If we were to move a charge from one point to another (for example, separating an electron from an atom), we would have to expend energy to do so.

The potential energy difference between these two points is referred to as

voltage.

It is denoted by the letter V (or sometimes E) and has units of volts, in honor of Alessandro Volta.

One volt is defined as one joule per coulomb.

$$1\text{ volt} \equiv 1\text{ joule} / 1\text{ coulomb}$$

DC Circuit Analysis

- Example 2: 100 joules are expended to move a 20 coulomb charge from point A to point B. Determine the resulting voltage.

DC Circuit Analysis

Power and Efficiency:

The terms power and energy are often used incorrectly as synonyms. Although related, they are not the same thing.

As already mentioned, energy is the ability to do work.

In contrast, power is the rate of energy usage.

Power is denoted by the letter P and has units of watts, although other units are sometimes used such as the horsepower (1 horsepower $\approx$ 746 watts).

One watt is defined as one joule of energy consumed per second.

$$1\text{ watt} \equiv 1\text{ joule} / 1\text{ second}$$

DC Circuit Analysis

Where

P = is the power in watts,

W = is the energy in joules,

t = is the time in seconds.

DC Circuit Analysis

- Example 3:

DC Circuit Analysis

Power can also be found by multiplying a current by the associated voltage. Note the definitions of current and voltage thus:

We already know that $P = W/t$, thus $P = IV$. This is known as power law.

$$P = I \times V$$

Where

P = is the power in watts,

I = is the current in amps,

V = is the voltage in volts.

DC Circuit Analysis

Efficiency

Efficiency is the ratio of useful output power to applied power expressed as a percentage.

It is denoted by the Greek letter η (eta) and is always less than 100%. E

Where

η = is the efficiency in percent,

P_{out} = is the output power,

P_{in} = is the input power.

Generally speaking, the higher the efficiency, the better. This implies less waste.

In other words, if a system is 30% efficient, then 70% of the input power is wasted, whereas if a system is 99% efficient, then only 1% of the input power is wasted.

DC Circuit Analysis

• Exercises:

1. An audio amplifier has a maximum rated output of 100 watts to a loudspeaker. If it exhibits an efficiency of 70%, determine the input power required and the amount of power wasted.
2. If a device draws 200 watts of power and has a useful output of 120 watts, determine the efficiency.

DC Circuit Analysis

Resistance and Conductance

Resistance is the characteristic of a material to oppose or resist the flow of current.

In other words, resistance is a measure of how difficult it is to establish a flow of current (i.e., “resistance to current flow”) under a given set of circumstances.

It is denoted by the letter R and has units of ohms, in honor of Georg Ohm, a researcher from the early 1800s.

The unit is denoted by the capital Greek letter omega, Ω .

DC Circuit Analysis

Sometimes it is more convenient to use the inverse view of resistance, and instead of referring to how difficult it is to establish a current, we would be interested in how easy it is to establish a current.

This is called conductance and it is the reciprocal of resistance.

Conductance is denoted by the letter G and has units of siemens, named after Werner von Siemens. The unit is abbreviated as S.

DC Circuit Analysis

The measure of a material's inherent and general ability to restrict current flow is called resistivity.

Resistivity is denoted by the Greek letter ρ (rho). All other factors being

equal, the higher the resistivity, the greater will be the resistance. Further, the greater the length of the material that the current must pass through, the greater the resistance. Finally, as the cross-sectional surface area grows, the resistance decreases.

Where

R = is the resistance in ohms,

ρ = is the resistivity,

l = is the length of the material,

A = is area of the face (h times w).

DC Circuit Analysis

DC Circuit Analysis

- Example 4:

DC Circuit Analysis

Various thicknesses of cables and wires are used for a wide variety of purposes.

To make this easier, wire thicknesses have been standardized. The most common standard in North America is the American Wire Gauge, or AWG. The larger the gauge number, the smaller the wire diameter, and the less current it safely can carry. To put the gauge numbers in perspective, typical small home appliances use 16 or 18 gauge wire, basic home wiring uses 12 gauge (with a 20 amp circuit breaker), and hook-up wire used in an electrical circuits or electronics laboratory solderless breadboard is commonly 22 or 24 gauge.

For general purpose wiring, copper is by far the most common metal used because it is highly conductive and relatively inexpensive. Some other metals are used in special cases, for instance, aluminum is used for long distance power transmission lines.

DC Circuit Analysis

DC Circuit Analysis

Resistors and the Resistor Color

Code:

Resistors are devices used to control the currents and voltages in a circuit. They are available in many shapes and sizes, and are normally designed to maintain stable ohmic values in spite of environmental changes such as temperature and humidity. As a general rule, the larger the resistor,

the more power it can handle.

DC Circuit Analysis

Resistors are linear bilateral devices. Being linear, their current-voltage relation can be drawn as a straight line. Bilateral means that the polarity of voltage or direction of current will not matter.

Resistors are available in standardized ohmic values and at standardized power ratings. Along with their resistance value, resistors also have a specified tolerance. This specifies an allowable range of variation of the stated value.

General purpose resistors use a color code to denote their value and tolerance. Typically, this will involve four color stripes: two for the precision/mantissa, one for the power of ten, and the fourth to indicate the tolerance. For higher precision, a five stripe version may be used with the first three denoting the precision/ mantissa. Alternately, high precision resistors may have their nominal value printed directly on them.

DC Circuit Analysis

DC Circuit Analysis

The tolerance band colors are as follows: For basic parts silver is $\pm 10\%$ and gold is $\pm 5\%$.

If the fourth band is omitted, this indicates a tolerance of $\pm 20\%$, although it is seldom used in modern designs.

For precision parts some colors are reused but follow the color code numerals: Brown is $\pm 1\%$ tolerance and red is $\pm 2\%$ tolerance.

Tighter tolerances are also available. Sometimes an extra band is added that indicates a reliability rating or temperature coefficient. In such cases, it is best to refer to the manufacturer's data sheets for details. We will not pursue these further.

DC Circuit Analysis

- Class work:

A precision resistor has

a color code of orange- Solution:

blue-green-brown-

The final band indicates that this brown. Determine its

is a $\pm 1\%$ tolerance component.

value and acceptable

The first four colors translate to range.

3, 6, 5 and 1. The value is “365 with 1 zero”, and thus 3650 ohms, or 3.65 k Ω . The allowable range is 3.65 k $\Omega \pm 36.5 \Omega$.

DC Circuit Analysis

- Class work:

Determine the nominal Solution:

value and tolerance of a

resistor with the color The first two colors translate to code green-blue-gold- 5 and 6. Gold in the third band silver. means “multiply by 0.1”. The final band indicates that this is a $\pm 10\%$ tolerance component.

Therefore, nominal value is 5.6

Ω . The allowable range is 5.6

$\Omega \pm 0.56 \Omega$.